

Below are three Kaggle capstone ideas, each with a short description and the datasets we'll use. I put in hyperlinks as well so you can see where the problem is from on Kaggle.

1. [Stable Diffusion - Image to Prompts](#)

Application:

I'll build a system that takes a Stable Diffusion 2.0 image and predicts the exact text prompt that generated it. First, I'll extract visual features with a neural network; then I'll feed those features into a decoder so it picks up style cues, composition, and keyword hints. For evaluation, following Kaggle, we'll compare the embeddings of predicted and true prompts using cosine similarity. A simple web demo will let anyone upload an image and see the model's best-guess prompt in real time.

Data: Thousands of AI-generated images paired with their exact text prompts from Kaggle's "Stable Diffusion - Image to Prompts."

2. [Google QUEST Q&A Labeling](#)

Application: I'll fine-tune a model on the Google QUEST dataset to score any question-and-answer pair on clarity, coherence, relevance, depth, and overall quality. We'll track regression accuracy on held-out examples and adjust prompts and loss weights to minimize big errors. The model will also generate a concise natural-language rationale for each score, making its judgments transparent and highlighting any systematic biases.

Data: Real Stack Exchange Q&A pairs with five continuous quality scores and rationale texts from Kaggle's "Google QUEST Q&A Labeling."

3. [Global Wheat Detection](#)

Application: I'll train an object detector on Kaggle's "Global Wheat Detection" images to spot and count every wheat head in a field photo. After tuning, I'll refine the box-filtering logic to prevent overlaps and integrate a simple counter that reports totals instantly. Finally, I'll wrap it in a friendly web demo where you upload a new image, see each head outlined, and get a head-count plus an estimated yield based on historical density stats.

Data: 3,300 high-resolution RGB images with COCO-style bounding boxes, plus 1,000 unlabeled test images from Kaggle's "Global Wheat Detection."